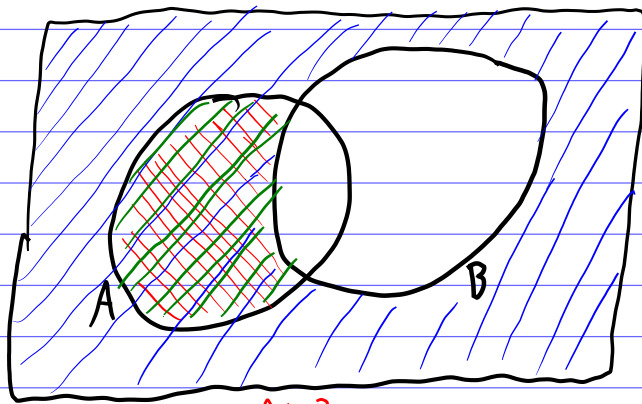


1 a



$A \cup \bar{B}$ & $A \setminus B$ $A \setminus B$ $\bar{B}$ $A \cap \bar{B}$
 cover the same area of the diagram & are thus equal

b $(C \setminus B) \setminus (B \setminus A) = C \setminus B$
 Using part a

$$(C \cap \bar{B}) \cap \overline{(B \cap \bar{A})} = C \cap \bar{B}$$

De Morgan's

$$(C \cap \bar{B}) \cap (\bar{B} \cup A) = C \cap \bar{B}$$

$$((C \cap \bar{B}) \cap \bar{B}) \cup ((C \cap \bar{B}) \cap A) = C \cap \bar{B}$$

absorption

$$(C \cap \bar{B}) = C \cap \bar{B}$$

2 double Inclusion

$$(A \cap B) \times C = (A \times C) \cap (B \times C)$$

$$LHS \subseteq RHS$$

$$x \in ((A \cap B) \times C)$$

$$x = \{(m, n) \mid m \in (A \cap B), n \in C\}$$

thus $m \in A$ & $m \in B$

so $x \in A \times C$ & $x \in B \times C$

Because both of the above are true

$$x \in (A \times C) \cap (B \times C)$$

thus the $LHS \subseteq RHS$

as both $LHS \subseteq RHS$ & $RHS \subseteq LHS$

$$LHS = RHS$$

$$RHS \subseteq LHS$$

$$x \in (A \times C) \cap (B \times C) \quad x = (x_1, x_2)$$

$$x \in (A \times C) = \{(a, b) \mid a \in A, b \in C\}$$

$$x \in (B \times C) = \{(a, b) \mid a \in B, b \in C\}$$

$$x_2 \in C$$

 $x_1 \in A$ & $x_1 \in B$ thus $x_1 \in (A \cap B)$

$$x \in \{(x_1, x_2) \mid x_1 \in (A \cap B), x_2 \in C\}$$

 $x \in ((A \cap B) \times C)$ by def of Cartesian Product
 thus $RHS \subseteq LHS$

3 8-10 chars
A-Z, a-z, 0-9 !?#&@%*~\$ _
2 from 3 of 4 16
total = 72

$P_8 =$ all 8 character valid passwords

Number of passwords with exactly 3 types

$$Q(a, b, c, k) = a \cdot b \cdot c \cdot (a+b+c)^{k-3}$$

a, b, c are number of options for each type, k is length

$u = \text{number of uppercase} = 26$

$$L = \text{number of lowercase} = 26$$
$$n = \text{Number of digits} = 10$$

$s = \text{number of special characters} = 10$

Passwords with all 4

$$|P_8| = Q(u, l, n, s) + Q(u, n, s, 8) + Q(u, l, s, 8) + Q(l, n, s, 8) + (u+l+n+s)^{8-4}$$

$$\approx 1.52737 \times 10^{13}$$

$$|P_8| + |P_9| + |P_{10}| = |P| \approx 6.025939 \times 10^{16}$$

4 | a

$$\begin{array}{ccccccc} \underline{1+1+1} & + & \underline{1+1} & + & \underline{1+1} & + & \underline{1+1} & + & \underline{1+1} & + & \underline{1+1} & + & \underline{1+1} & + & \underline{1+1} & + & \underline{1+1} & + & \underline{1+1} \\ x_1 = 3 & & x_2 = 2 & & & & x_3 = 2 & & & & x_4 = 2 & & & & x_5 = 2 & & & & x_6 = 2 \end{array}$$

Placing 3 dividers into the above

$$\binom{23}{3} = \frac{23!}{3! \cdot (20)!}$$

b # no longer counted; i.e. w/ at least 1 x > 9

$1+1+1+1+1+1+1+1+1+1+1+1+1+1+1+1+1+1$

to ensure at least 1 x 210
still placing 3 dividers

$$\binom{13}{3} = \frac{13!}{3!(10!)}$$

$$\frac{23!}{3! \cdot (20)!} - \frac{13!}{3! \cdot (10)!}$$